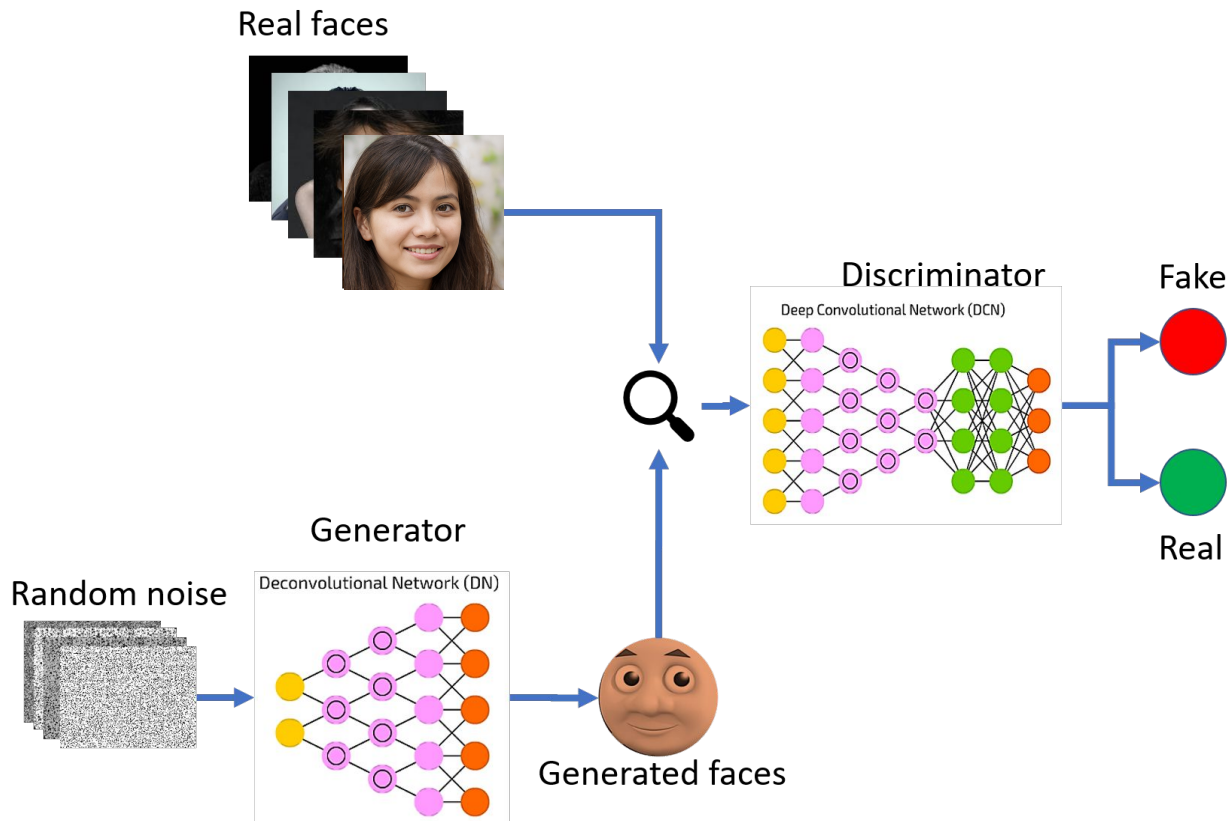


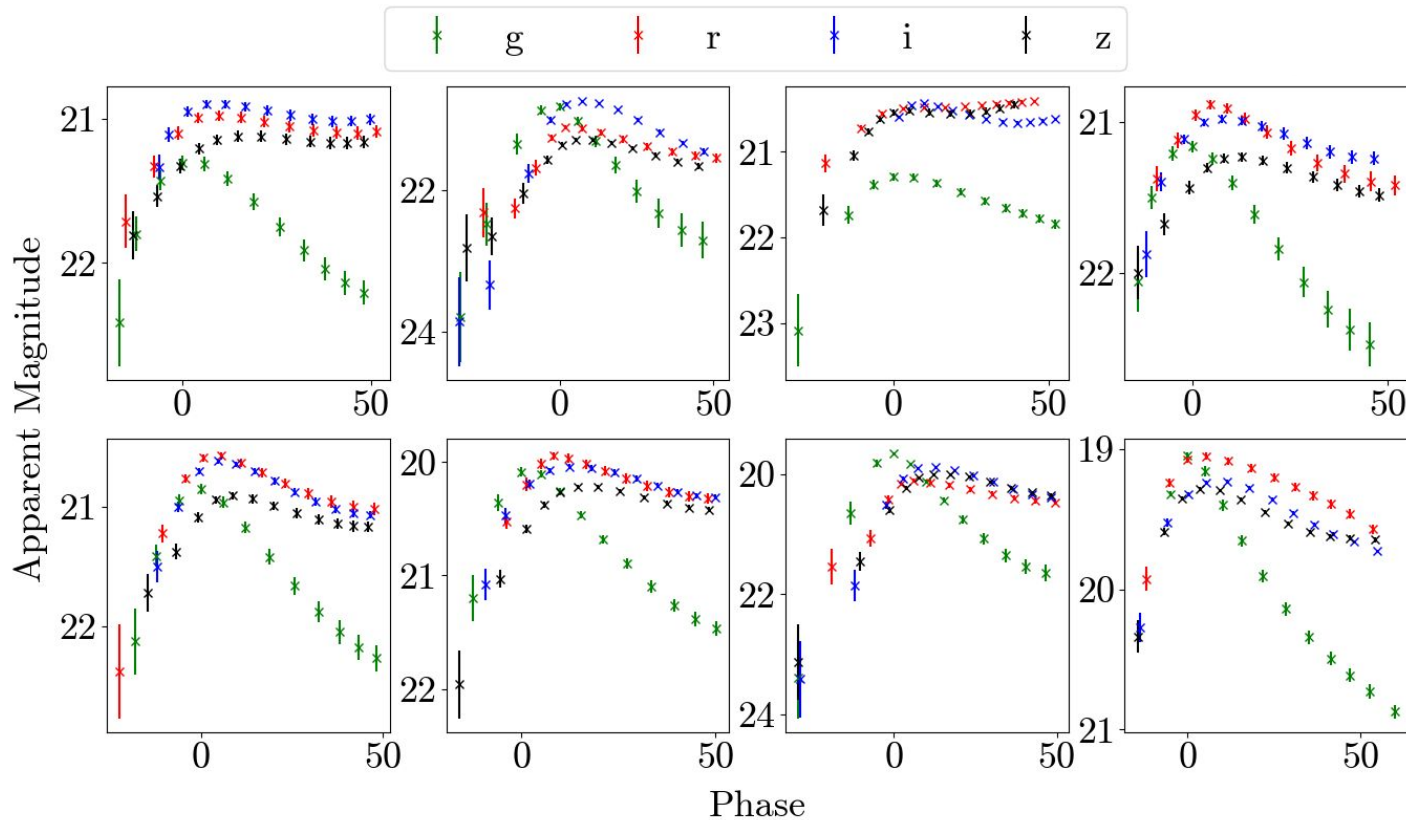
Augmenting Supernova Training Sets Using Generative Adversarial Networks

Matt Grayling

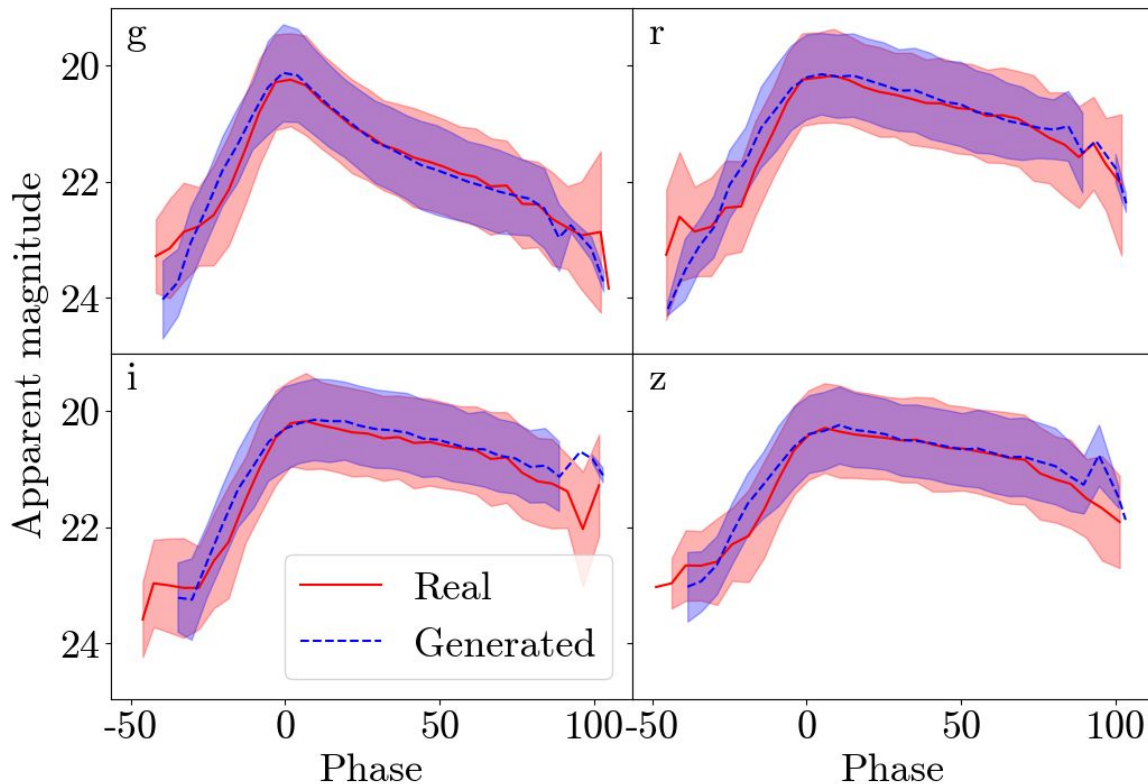
Generative Adversarial Networks



Generated Light Curve Examples



Training vs generated data comparison



Conclusions

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- Generative Adversarial Networks (GANs) are a powerful data augmentation tool which have been applied in other fields to improve classifiers
- It is possible to generate physically realistic SN light curves using GANs
- These techniques can be used to help improve the performance of photometric classifiers